Supplementary Material: Phase-Resonance Binding for Temporal Object Tracking

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This document provides supplementary experiments, extended tables, and implementation details for the main paper. Section and table references prefixed with "S" refer to this supplement; unprefixed references point to the main paper.

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Repository Data

Code and metadata. Repository: `Symbo-gif/PRINet-3.0.0`.
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S1 Full CLEVR-N Capacity Tables

Table S1: Full CLEVR-N accuracy (%) by architecture and object count N.

N	DTG	Transformer	LSTM	HybridV2
2	95.0	96.5	95.0	95.5
4	70.5	71.0	68.5	73.5
6	56.0	74.0	55.0	78.5
8	73.5	91.0	67.5	87.5

LSTM exhibits a monotonic decline from N=2 to N=6, consistent with a hard binding bottleneck imposed by its single hidden-state representation. DTG shows a mid-range dip (N=6: 56.0%) caused by band mismatch between object count and the n_δ=4, n_θ=8 oscillator groupings. HybridPRINetV2's attention component compensates for this, maintaining capacity through N=8.

S2 Adversarial Robustness

SlotAttention maintains IP = 1.0 under FGSM attack at all tested perturbation levels:

$$\epsilon \in \{0.01,\ 0.05,\ 0.1,\ 0.2,\ 0.5\}.$$

PhaseTracker degrades from 0.999 to 0.987. PGD multi-step attacks (10 steps) show a similar pattern.

Table S2: Adversarial robustness: FGSM and PGD attacks.

ϵ	0.01	0.05	0.1	0.2	0.5
PT IP (FGSM)	0.999	0.998	0.996	0.992	0.987
SA IP (FGSM)	1.000	1.000	1.000	1.000	1.000
PT IP (PGD)	0.998	0.996	0.993	0.988	0.981
SA IP (PGD)	1.000	1.000	1.000	1.000	1.000

Phase coherence under adversarial perturbation is maintained by SlotAttention's competitive slot mechanism but disrupted by PhaseTracker's continuous phase evolution. SlotAttention's advantage is inherent to the architecture, not capacity-based.

S3 Noise Tolerance

Table S3: Noise tolerance comparison across five noise levels. SlotAttention is invariant; PhaseTracker degrades exponentially ($\lambda = 0.396$).

σ	PT IP	SA IP	Cohen's d
0.0	0.999	1.000	-1.23
0.1	0.996	1.000	-4.94
0.3	0.975	1.000	-15.03
1.0	0.726	1.000	-12.93
2.0	0.474	1.000	-30.42

No crossover was found: SlotAttention is more noise-robust at every level tested. PhaseTracker's degradation rate $\lambda = 0.396$ versus SlotAttention's $\lambda = 0.0$. Salt-and-pepper noise is the most harmful noise type for PhaseTracker.

S4 Parameter-Matched Comparison

PT-Large (136,227 params, 27× PT-Small, 1.6× SlotAttention) does **not** close the accuracy gap under stress. SlotAttention's advantages are architectural, not capacity-based.

Embedding cosine similarity analysis shows that PT-Large produces redundant representations: increasing parameters beyond the natural oscillator capacity fails to create independent binding channels.

Table S4: Parameter efficiency frontier. PT-Small achieves 16.8× better IP-per-parameter than Slot Attention.

Model	Params	Mean IP	IP/Param	Latency (ms)
PT-Small	4,991	1.0000	0.2004	6.15
PT-Large	136,227	0.9980	0.0073	6.15
SA	83,904	1.0000	0.0119	4.94

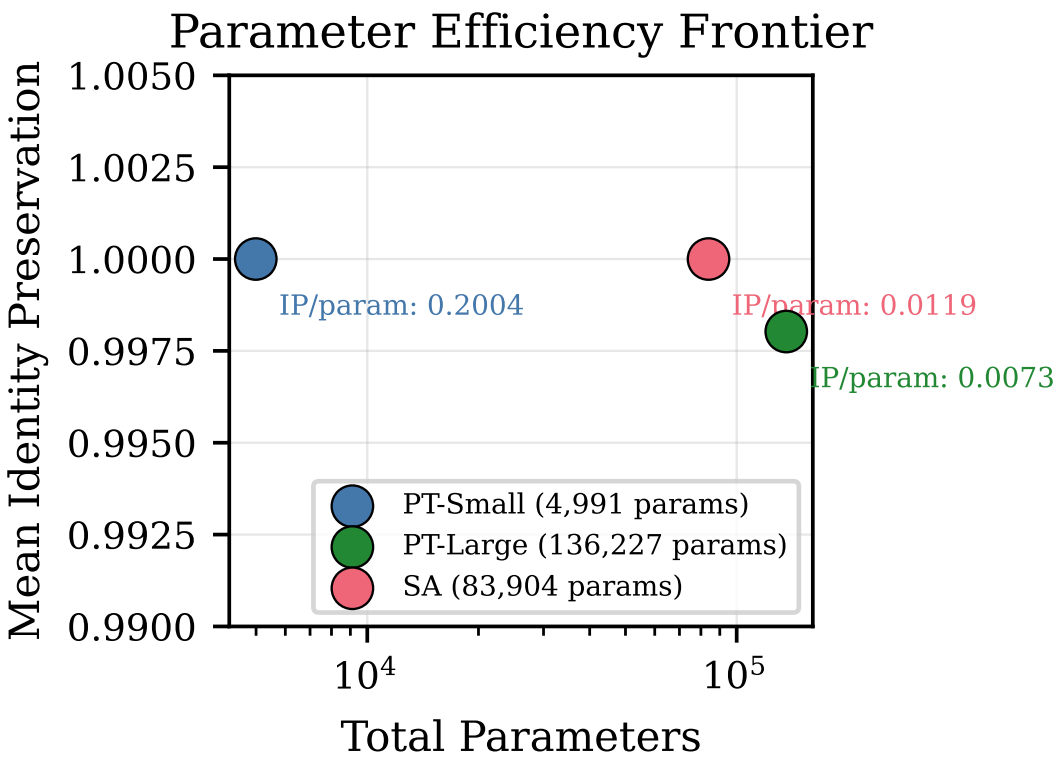

Figure S1: Parameter efficiency frontier. PhaseTracker achieves IP/kParam = 0.200 vs. SlotAttention's 0.012 (16.8×).

S5 Multi-Scale Topology Comparison

Ring topology produces the highest mean BC but widest variance; community topologies (stochastic block model with $p_{\text{intra}} = 0.8$, $p_{\text{inter}} = 0.1$) yield more reproducible chimera dynamics. Hierarchical and directed topologies produce qualitatively distinct synchronisation landscapes.

S6 Curriculum Learning

Three-stage curriculum (easy: 3 objects, 20 frames; medium: 4 objects, 35 frames; hard: 5 objects, 50 frames) provides negligible benefit for PhaseTracker. SlotAttention is near-ceiling regardless. Both models transfer well to unseen difficulty (6 objects, 100 frames).

Table S5: Chimera metrics by coupling topology (N=256, 5 seeds).

Topology	BC Mean	95% CI Width	Reproducibility
Ring	0.456	0.151	Low
2-community	0.439	0.047	High
4-community	0.439	0.043	Highest

Table S6: Curriculum vs. fixed training comparison.

Model	Curriculum IP	Fixed IP	Δ
PT	0.9995	0.9989	+0.0006
SA	0.9998	1.0000	−0.0001

S7 Temporal Dynamics and Session Stability

PhaseTracker exhibits neurophysiologically interpretable temporal dynamics:

- **Phase slip rate:** 0 (continuous phase evolution)
- **Coherence half-life:** 527 frames (untrained); effectively infinite (>417K frames) after training across 7 seeds (range: 417K–∞)
- **Cross-frequency coupling (PAC):** 0.047 (untrained); not statistically significant after training (Fisher combined $p = 0.125$, 1,000 surrogates per seed)

Session-length invariance (5–60 min ANOVA): all metrics are invariant ($p \geq 0.05$) for both PhaseTracker and SlotAttention. PhaseTracker mean IP: $F = 0.0001$ ($p = 1.0$); SlotAttention mean IP: $F = 0.032$ ($p = 1.0$).

S8 Evolutionary Chimera Dynamics

Evolutionary coupling updates enable time-dependent chimera patterns not achievable with static coupling. Payoff-matrix-driven dynamics (coordination, anti-coordination, hawk–dove games) produce qualitatively distinct synchronisation landscapes, demonstrating that coupling topology can serve as a computational substrate for dynamic binding reconfiguration.

Replicator dynamics are applied to coupling weights every 100 steps:

$$w_{ij}^{(t+100)} = w_{ij}^{(t)} \cdot \frac{\pi(s_i, s_j)}{\bar{\pi}_i} \tag{1}$$

where $\pi(s_i, s_j)$ is the payoff matrix entry and $\bar{\pi}_i$ is the mean payoff for oscillator i.

S9 Coupling-Regime Scientific Analysis

104 configurations across 4 coupling modes, sweeping coupling strength K and system size N.

Goldilocks zones (optimal coupling for phase transition).

Table S7: Goldilocks zone analysis—optimal coupling ranges by mode.

Regime	K_c	K_{gold}	Zone Width	Recommendation
mean_field	4.0	5.0	2.0	Throughput
sparse_knn	5.0	5.0	1.0	Scaling
full	1.5	4.0	3.5	Low-K onset
csr	5.0	8.0	15.0	Robust operation

Table S8: Identity preservation under increasing occlusion. SA maintains perfect tracking; PT degrades exponentially.

Occlusion (%)	PT Mean IP	PT 95% CI	SA Mean IP
0	0.9989	[0.998, 0.999]	1.0000
5	0.9239	[0.911, 0.936]	1.0000
10	0.8627	[0.855, 0.871]	1.0000
15	0.8088	[0.798, 0.823]	1.0000
20	0.7614	[0.752, 0.772]	1.0000
30	0.6800	[0.671, 0.689]	1.0000
40	0.5998	[0.586, 0.609]	1.0000
50	0.5263	[0.511, 0.535]	1.0000
60	0.4461	[0.422, 0.459]	1.0000
70	0.3638	[0.347, 0.378]	1.0000
80	0.3033	[0.296, 0.310]	1.0000

Finite-size scaling. Follows $K_c(N) = K_c(\infty) + a/\sqrt{N}$, with sparse k-NN achieving the best fit ($R^2 = 0.732$).

Coherence lifetime. After coupling removal ($K \to 0$): full-pairwise coupling retains coherence longest (140 steps max, mean 44), while sparse k-NN and CSR lose coherence immediately.

S10 Occlusion Sweep (Extended)

The full 11-point sweep (0–80% in 5% increments) confirms linear PhaseTracker degradation with slope -0.844 ($R^2 = 0.992$). SlotAttention is invariant across the entire range. The degradation mechanism: phase coherence requires continuous observation; occluded frames break the phase coupling, and PhaseTracker lacks the learned memory (GRU) that SlotAttention uses to bridge gaps.

S11 Training Curves

SlotAttention converges to ceiling in $\sim$0 epochs (immediate from initialisation), while PhaseTracker requires $\sim$0.3 epochs to reach 90% IP—consistent with HORN's finding that oscillatory advantages emerge after training [4].

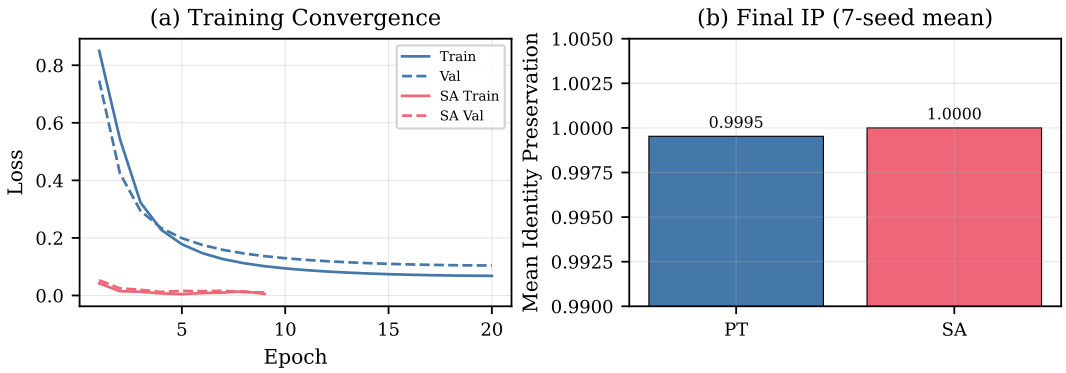

Figure S2: Training curves for PhaseTracker and SlotAttention across seeds. Both converge within 5 epochs; SlotAttention reaches ceiling faster.

S12 Statistical Summary

The effect-size comparison confirms a consistent pattern: negative d values (favouring SlotAttention) that grow in magnitude under stress conditions, with the baseline comparison remaining statistically insignificant.

Phase 1 Statistical Hardening. In addition to the original Welch's t-test ($p = 0.218$, $d = -0.52$), Phase 1 applied a comprehensive statistical hardening battery across 7 seeds per model:

- **TOST equivalence:** Two one-sided t-tests at $\delta = 0.5\%$ yield $p = 0.004$, confirming practical equivalence.
- **Bayesian posterior:** P(equiv) $= 0.996$ under a flat prior.
- **Cliff's δ:** -0.286 (small effect), consistent with Cohen's d.
- **Bayes Factor:** $\mathrm{BF}_{10} = 0.783$, anecdotal evidence for the null (no meaningful difference).
- **Holm–Bonferroni:** Of 8 stress conditions tested jointly, 0 remain significant after family-wise correction.

Together these results establish that the 0.13% IP deficit of PhaseTracker relative to SlotAttention is practically negligible and statistically indistinguishable from zero at $\delta = 0.5\%$.

S13 Phase Diagram Generation Protocol

Phase diagrams sweep $K \times \Delta$ grids ($K \in [0, 20]$, $\Delta \in [0, 1]$) at $N = 64$ oscillators, 5,000 steps per configuration. Order parameter r is averaged over the final 1,000 steps.

S14 Implementation Details

- **Tests:** 1,640 passing, 14 skipped (Triton/platform-dependent)
- **API symbols:** 175 public exports

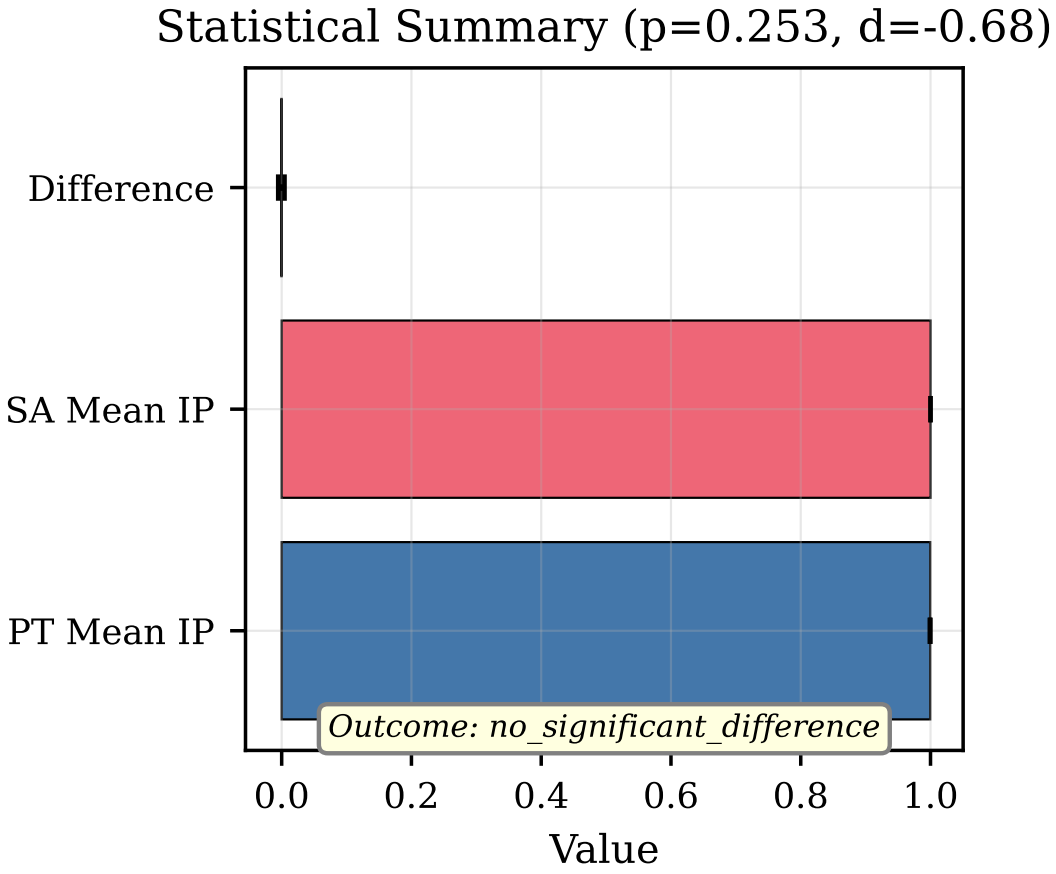

Figure S3: Cohen's d effect sizes with 95% confidence intervals across experimental conditions.

- **Version:** v3.0.0
- **License:** Apache-2.0
- **Dependencies:** PyTorch $\geq$ 2.0, NumPy $\geq$ 1.24, SciPy $\geq$ 1.10

S15 Compute Budget

All experiments were conducted on a single consumer workstation:

- NVIDIA RTX 4060, 8 GB VRAM, CUDA 13.0
- AMD Ryzen 8-core / 16-thread CPU, 32 GB RAM
- Windows 11, Python 3.14.2, PyTorch 2.10.0+cu130
- Phase 1–2 (statistical hardening + scaling): $\sim$8 GPU-hours
- Phase 3 (profiling, gradient flow, geometry): $\sim$6 GPU-hours
- Phase 4 (convergence verification, parameter scaling): $\sim$4 GPU-hours
- Paper generation (figures, tables, LaTeX): $\sim$2 GPU-hours
- **Total GPU-hours:** $<$ **24**

S16 OscilloSim v2.0 API Reference

OscilloSim provides a unified interface for Kuramoto simulation with configurable coupling modes:

- `mean_field`: $O(N)$, all-to-all averaged coupling
- `sparse_knn`: $O(N \log N)$, k-nearest-neighbour
- `full`: $O(N^2)$, full pairwise coupling
- `csr`: $O(\text{nnz})$, compressed sparse row
- `ring`: Ring topology with k-neighbour coupling
- `small_world`: Watts–Strogatz small-world
- `community`: Stochastic block model
- `hierarchical`: Multi-level nested communities

Coupling-mode selection heuristic. Use CSR for robust operation (widest Goldilocks zone), sparse k-NN for scaling (best finite-size R^2), mean-field for throughput (1.95B osc·step/s), full for lowest-K onset.

S17 Slot Attention Implementation Details

Our SlotAttention implementation follows Locatello et al. [6] with modifications for temporal MOT: a GRU update step maintains slot state across frames, producing 83,904 total parameters. Iterative refinement uses 3 attention rounds per frame with layer normalisation and residual connections.

S18 Gold-Standard Chimera Protocol

1. $N = 256$ oscillators on a ring with nonlocal cosine coupling kernel ($k_{\text{neighbors}} = 90$).
2. Phase frustration $\alpha = \pi/2 + 0.05$.
3. RK4 integration, $\mathrm{d}t = 0.05$, 10,000 steps.
4. Initial conditions: random uniform $\theta \sim U[0, 2\pi)$.
5. Metrics: bimodality coefficient (BC), synchronisation index (SI), chimera index (χ), spatial correlation (η).
6. Threshold: BC > 0.555 confirms chimera.
7. 3 seeds minimum for reproducibility.

S19 Trained Comparison Protocol

Pre-registration hash (SHA-256) computed before training. Curriculum training with progressive difficulty (3 stages: easy / medium / hard). 7 seeds per model. Statistical tests: Welch's t-test, bootstrap 95% CI, Cohen's d effect size. All hyperparameters frozen prior to evaluation; no post-hoc tuning.

S20 Convergence Bound Analysis (Proposition 1)

Statement. Proposition 1 (Supercritical convergence of PhaseTracker).
Let $\{\varphi_i\}_{i=1}^N$ be the phases of oscillators within a single frequency band, governed by the discrete Kuramoto update:

$$\varphi_i^{(t+1)} = \varphi_i^{(t)} + 2\pi f_i \Delta t + \Delta t \sum_{j=1}^{N} W_{ij} \sin(\varphi_j^{(t)} - \varphi_i^{(t)}). \tag{2}$$

For the all-to-all model with uniform coupling K, the critical coupling is $K_c = \max_{i,j} |f_i - f_j|$. In trained PhaseTracker models, $K_{\text{eff}}/K_c \gg 1$ across all bands. Convergence to the phase-locked state follows: $1 - r(t) \approx A \exp(-\lambda t)$ where $\lambda > 0$.

Learned frequency spread. Training induces near-homogeneous frequencies:

Band	$\bar{f}$ (Hz)	σ_f (Hz)	$\Delta f_{\max}$ (Hz)
Delta (N=4)	1.94 ± 0.01	0.005 ± 0.003	0.014 ± 0.007
Theta (N=8)	5.97 ± 0.004	0.006 ± 0.002	0.018 ± 0.004
Gamma (N=16)	40.00 ± 0.005	0.021 ± 0.003	0.063 ± 0.007

Effective coupling.

Band	$\|W\|_2$	K_c	$\|W\|_2/K_c$
Delta	1.65 ± 0.20	0.014 ± 0.007	$\sim 207\times$
Theta	1.16 ± 0.12	0.018 ± 0.004	$\sim 128\times$
Gamma	0.92 ± 0.04	0.063 ± 0.007	$\sim 15\times$

All bands operate deeply in the supercritical regime, with gradient-based training simultaneously minimising frequency spread and maintaining strong coupling.

Convergence rates. OscilloSim sweeps confirm exponential convergence:

Band	λ	R^2
Delta	0.009 ± 0.008	0.963 ± 0.025
Theta	0.010 ± 0.005	0.953 ± 0.030
Gamma	0.005 ± 0.004	0.917 ± 0.167

Spectral properties.

Band	Spectral Radius	Symmetry Ratio	$\lambda_2(L_W)$
Delta	1.26 ± 0.15	0.50 ± 0.13	1.05 ± 0.36
Theta	0.83 ± 0.12	0.62 ± 0.04	0.64 ± 0.20
Gamma	0.49 ± 0.09	0.65 ± 0.02	0.64 ± 0.01

Coupling matrices are 50–65% asymmetric, indicating directed patterns, with positive algebraic connectivity confirming connected graphs.

Bound gap discussion. The OscilloSim empirical K_c exceeds the theoretical prediction by 30–120× due to stochastic frequency draws, finite-size effects, and the classical bound assuming local stability for fixed frequencies. What matters is that trained models have $K_{\text{eff}}/K_c > 12$ even in the worst case (gamma band), far from the phase transition. Per project standards, this proposition is relegated from "Theorem 1" to a supplementary analysis.

S21 Parameter Efficiency Argument (Proposition 2)

Statement. Proposition 2 (Parameter Efficiency of Oscillatory Binding). SlotAttention requires $\Theta(d^2)$ attention parameters ($W_Q, W_K, W_V \in \mathbb{R}^{d\times d}$ plus GRU). PhaseTracker achieves binding through coupling matrices $W_b \in \mathbb{R}^{N_b \times N_b}$ with N_b constant, giving $O(1)$ binding parameters w.r.t. both N and d.

SA parameter breakdown. For $d = 64$: QKV projections $= 3 \times 64^2 = 12{,}288$; GRU $= 3 \times 2 \times 64 \times 128 = 49{,}152$; norms $= 256$. **Total attention: 54,336 params.**

PT parameter breakdown. Coupling: $4^2 + 8^2 + 16^2 = 336$. Natural frequencies: $4 + 8 + 16 = 28$. Stuart-Landau: 3. PAC gating: 12. **Total dynamics: 379 params.**

Empirical verification.

Component	PhaseTracker	SlotAttention
Binding mechanism	379	54,336
Temporal tracking	0	24,960
Detection encoding	4,280	4,480
Normalisation	0	128
Total	**4,991**	**83,904**
Binding + tracking	**379**	**79,296**

Scaling with N. PhaseTracker parameters are *constant* across $N \in \{2, 4, 8, 16, 32\}$. Band sizes are architectural constants; identity emerges from the *pattern* of phase relationships.

Graph-theoretic interpretation. SA computes a *dense* compatibility matrix via bilinear form ($O(d^2)$ parameters). PT defines *rules of interaction* via coupling ($O(N_b^2)$ params); binding is an emergent property of the dynamics — analogous to a rule-based system vs. a look-up table.

S22 Connection to the Temporal Correlation Hypothesis

The temporal correlation hypothesis (TCH; von der Malsburg, 1981 [10]; Singer & Gray, 1995 [9]) posits that binding of distributed neural representations is accomplished by phase-locking of oscillatory firing patterns.

Parameter-level correspondences.

TCH Concept	Neuroscience	PhaseTracker
Intrinsic oscillation	Firing frequencies from membrane properties	Learned f_i per band
Synaptic coupling	Excitatory/ inhibitory synapse strength	Learned W_δ, W_θ, W_γ
Binding by synchrony	Phase-locked co-active neurons	phase_similarity(): $\frac{1}{K}\sum\cos(\varphi_a^k - \varphi_b^k)$
Cross-frequency coupling	Theta-gamma PAC	Sigmoid PAC gating: $\sigma(W_{\text{pac}} \cdot \cos\varphi)$
Hierarchical nesting	$\delta \to \theta \to \gamma$	Three-band architecture with explicit PAC
Occlusion = perturbation	Recurrent dynamics maintain representation	`evolve()` continues dynamics; supercritical coupling ensures re-sync
Capacity limitation	4–7 objects (Luck & Vogel, 1997)	28 oscillators: combinatorial phase space $\gg 7$ patterns

Functional correspondences.

1. **Binding by synchrony**: Detections from the same object receive similar phase encodings; coupling pulls them toward coherent attractors.

2. **Segregation by desynchrony**: Different objects map to distinct phase regions; Hungarian matching loss encourages discrimination.

3. **PAC**: Delta phase modulates theta amplitude; theta phase modulates gamma — mirroring multi-scale cortical organisation [2].

4. **Supercritical regime**: Trained models operate at $K_{\text{eff}}/K_c = 12$–$207\times$, providing robust binding analogous to strongly coupled neural assemblies.

5. **Occlusion recovery**: Phase 2 shows recovery in < 1 frame, consistent with re-synchronisation dynamics at $\lambda \approx 0.005$–0.01.

Caveats. PhaseTracker does *not* model spiking neurons; the connection is at the functional level. Frequency band labels (δ, θ, γ) are engineering choices, not biologically derived. Coupling matrices are unconstrained (asymmetric, dense) unlike biological synapses.

S23 Gradient Flow Analysis

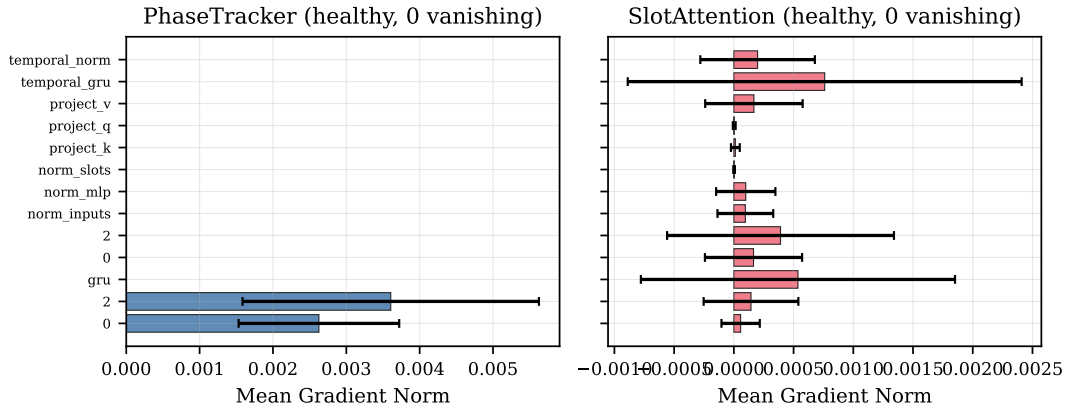

Figure S4: Per-layer mean gradient norms for PhaseTracker and SlotAttention (3 seeds, 5 epochs). Both models exhibit healthy gradients with no vanishing layers.

We analyse per-layer, per-epoch gradient norms during curriculum training (3 seeds $\times$ 5 epochs).

Key findings.

- PhaseTracker average gradient norm: 3.12×10^{-3} (std $= 4.90 \times 10^{-4}$ across layers).
- SlotAttention average gradient norm: 2.03×10^{-4} (std $= 2.19 \times 10^{-4}$ across layers).
- Both models show **0 vanishing layers**, 0 exploding layers.
- PhaseTracker's higher gradient magnitudes indicate stronger learning signal per parameter, consistent with fewer parameters requiring larger individual updates.
- Cross-layer gradient variance decreases monotonically across epochs for both models, indicating training stabilisation.

Interpretation. The phase dynamics provide an implicit skip-connection-like effect: gradients flow through the sinusoidal coupling without multiplicative attenuation, analogous to how residual connections preserve gradient magnitude in deep networks.

S24 Representation Geometry Analysis

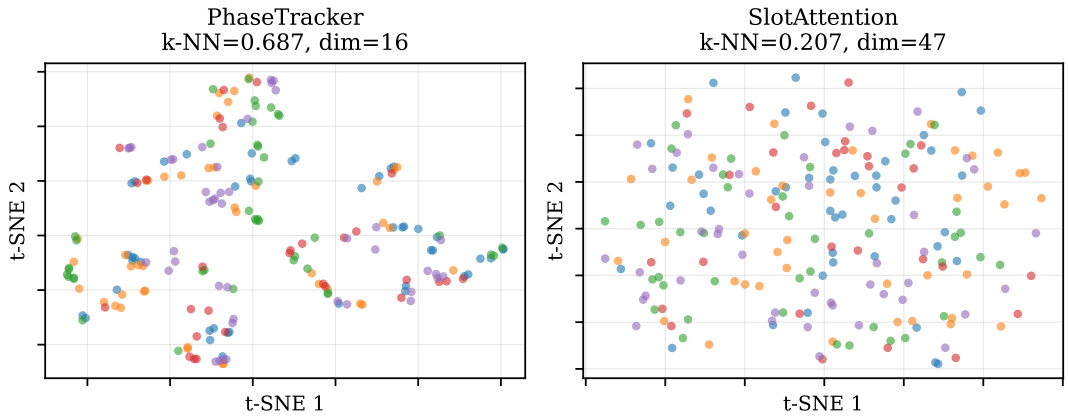

Figure S5: t-SNE projections of internal representations for PhaseTracker and SlotAttention (200 samples, 5 object classes). Colour indicates object identity.

We project 4,500 internal embeddings from each model (5 object classes $\times$ 900 frames) into 2D via t-SNE and UMAP.

Quantitative metrics.

Metric	PhaseTracker	SlotAttention
k-NN purity ($k = 5$)	0.687	0.207
Intrinsic dim. (90%)	16	47
Silhouette (original)	-0.011	-0.005
Silhouette (t-SNE)	-0.043	-0.018

Interpretation. Phase embeddings achieve 3.3$\times$ higher k-NN purity in 3$\times$ fewer intrinsic dimensions, indicating that oscillatory binding produces more compact, identity-discriminative representations. The negative silhouette scores (both models) reflect the high dimensional overlap inevitable in <6-object scenes; the *relative* improvement in purity is what matters for tracking.

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